Huanxuan Liao

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Education

Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing **2023 – Present**

PhD student in Pattern Recognition and Intelligent Systems (NLPR). Advisors: Shizhu He and Kang Liu

North China Electric Power University (NCEPU), Beijing **2019 – 2023**

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Ranked 1/13

Research Interests

I work toward **foundation models for long-horizon adaptive intelligence**: models that use growing contexts effectively, retain reusable capabilities, and act reliably over extended interactions.

- **Long-Context Data & Model Co-Design**: long-context data curation and synthesis, length- and capability-aware mixtures, scalable pretraining and mid-training, efficient attention, KV-cache optimization, and architectures to jointly improve long-context capability, scalability, and efficiency (LCLM-Horizon, HyCo$_2$, SparK, ResAdapt, AllocEmbed).
- **Knowledge & Continual Adaptation**: acquiring, transferring, internalizing, and updating knowledge and capabilities across evolving tasks, while balancing plasticity, retention, and generalization (TAGI, AAG, NesyCD, SKIntern, SpaRTA, SHIP, DyPRAG).
- **Agent Learning & Self-Improvement**: persistent memory, long-horizon decision-making, and learning from interaction and feedback to verifiably refine agent memory, policies, and learning procedures (CITI, KATE, OPERATE, Distill Yourself, SLE, BuPO, Tool-RL-Box).

Research Experience

Tencent (Project UP) Beijing **Jul. 2026 – Present**

Foundation Model Pretraining and Efficient Inference

- **Long-context data and training**: Designed multi-QA MRCR synthesis with long-range dependencies, training-signal monitoring, and **10B-token ablations** of QA structure and mixtures. Earlier matched-mixture MRCR: **0.854** → **0.911**; delivered **290B tokens across eight 8K–1M ranges**.
- **Efficient attention**: Shared low-dimensional indices, block-importance training, parameter folding, and fused operators reduced Select cost. Archived MiniCPM/H200 tests (B4, 32K–128K): **1.11–1.36**× **Prefill speedup over Sparse**; RULER four-length mean **75.08** → **76.33**; LongBench **45.06** (Dense: 44.78).
- **SVG pretraining**: Built **8.13M samples (41B tokens)** across static graphics, chemistry, and animation, using geometric standardization, image–code validation, and cross-corpus perceptual deduplication. Applied **Qwen3-VL-Embedding coarse screening and SVGFaith-Score fine screening** with human-calibrated thresholds.
- **Sketch-to-web code**: Synthesized layout-preserving styles using **DOM wireframes, Canny edges, and text/image placeholder replacement**. Visual filtering yielded **48K sketch–code pairs**; built a human-drawn evaluation set and dual-judge quality scoring.

ByteDance Beijing **Jun. 2025 – Jun. 2026**

Multimodal Understanding

- **AIGC detection**: Deployed efficient multimodal features for TikTok AIGC detection, improving online F1 by >**3%** and reducing compute by **30%–50%**; reused these features for embedding retrieval.
- **Efficient video reasoning**: Proposed **ResAdapt** and CAPO to adjust each frame's input resolution. Reduced visual tokens by >**90%**, processed up to **16**× more frames at the same compute, and achieved a >**15%** relative gain in long-video reasoning.
- **Efficient video retrieval**: Contributed to **AllocEmbed**, which uses low-resolution previews to allocate frame resolution and retrieval-driven policy optimization (RDPO) to train the allocator. At matched visual-input budgets, LongRet Hit@1 improved by **5.69–9.09 percentage points** across four frozen backbones (four-subset averages); the allocator transferred without retraining.
- **Model efficiency**: Compared visual-token and KV-cache compression to measure their accuracy and compute costs; explored layer skipping and dynamic boundaries as further efficiency options.
- **Agent evaluation**: Built **OPERATE** to test when agents should gather information, act, or keep a plan as their environment changes. Scored decisions using execution records and comparisons of alternative actions under the same simulation seed.

Ant Group Beijing **Feb. 2025 – May 2025**

Long-Context Reasoning

- **Long-context compression**: Proposed HyCo$_2$ to preserve a document's overall meaning and relevant evidence in a shorter input. Matched uncompressed performance with about **88.8%** fewer input tokens; achieved a **13.1% average relative gain** across seven knowledge-intensive QA tasks.
- **Survey and open source**: Core contributor to the LCLM-Horizon survey, organizing research on data, models, training, and evaluation; curated and maintained Awesome-LCLM (**Curator & Maintainer**).

Selected Publications [full list]

Long-Context Data & Model Co-Design

[ACM TOIS'26] **Huanxuan Liao**, Yao Xu, Shizhu He, Jun Zhao, Kang Liu. HyCo$_2$: Hybrid Context Compression with Global Semantics and Local Evidence.

[arXiv'26] **Huanxuan Liao**, Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, Kang Liu. ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning.

[arXiv'26] Song Jin, Zhongtao Jiang, Chenglei Shen, **Huanxuan Liao**, Haozhe Chi, Zhiwei Wang, Kun Xu, Yong Liu. Allocate Before You Embed: Adaptive Visual Input Allocation for Video Embeddings.

[AAAI'26] **Huanxuan Liao**, Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, Kang Liu. SparK: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning.

[arXiv'25] Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, **Huanxuan Liao**, Haoran Que, Zekun Wang, et al. A Comprehensive Survey on Long Context Language Modeling. **(Core Contributor)**

Knowledge & Continual Adaptation

[ACL'26] **Huanxuan Liao**, Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, Kang Liu. Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs.

[ACL'26] Wangtao Sun, Haotian Xu, **Huanxuan Liao**, Xuanqing Yu, Zhongtao Jiang, Shizhu He, Jun Zhao, Kang Liu. Shuttle Between Symbolic Instructions and Neural Parameters of Large Language Models.

[NeurIPS'26] Yuqiao Tan, Shizhu He, **Huanxuan Liao**, Jun Zhao, Kang Liu. Dynamic Parametric Retrieval Augmented Generation for Test-time Knowledge Enhancement. *TTCL Workshop*.

[AAAI'25] **Huanxuan Liao**, Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Jun Zhao. Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks.

[COLING'25] **Huanxuan Liao**, Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, Kang Liu. SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models.

[COLING'25] **Huanxuan Liao**, Shizhu He, Yao Xu, Yuanzhe Zhang, Shengping Liu, Kang Liu, Jun Zhao. Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering.

[NeurIPS'24] **Huanxuan Liao**, Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, Jun Zhao. From Instance Training to Instruction Learning: Task Adapters Generation from Instructions.

Agent Learning & Self-Improvement

[EMNLP'26] Yuqiao Tan, Minzheng Wang, Shizhu He, **Huanxuan Liao**, Chengfeng Zhao, Qiunan Lu, Tian Liang, Jun Zhao, Kang Liu. Bottom-up Policy Optimization: Your Language Model Policy Secretly Contains Internal Policies. **Oral.**

[ICML'26] Yupu Hao, Zhuoran Jin, **Huanxuan Liao**, Kang Liu, Jun Zhao. Why Multi-Step Tool-Use Reinforcement Learning Collapses and How Supervisory Signals Fix It. *FAGEN Workshop*.

Awards and Scholarships

CAST Young Science and Technology Talent Cultivation Program 2026
Doctoral Student Special Plan
National Scholarship ×3 2021 / 2022 / 2025
Beijing Outstanding Graduate 2023.06

Professional Services

Conference Reviewing: ACL ARR 2024–2026. NeurIPS 2025–2026. ICML 2026. COLM 2026. ICLR 2025–2026. AAAI 2026.
Open Source: Awesome-LCLM: Curator & Maintainer; NVIDIA/kvpress: Contributor; Distill Yourself: Core Logic & Code Contributor.